SalesAgents AI - AI Intern (Prompt Engineer) Assignment Prep Guide Ref Material

The assignment asks you to build an AI Voice Agent for Home Credit that informs existing customers about a pre-approved Loan Against Property (LAP) offer, collects **7 eligibility data points sequentially**, handles interruptions/diversions, immediately disqualifies ineligible customers, detects loan-transfer cases, and finally hands qualified leads to a senior loan expert.

1. Difficulty Rating

Overall Difficulty: 7.5/10 — Intermediate to Advanced

Area	Difficulty
Understanding requirements	★★☆☆☆
Basic prompt writing	★★☆☆☆
Conversation-flow design	★★★★☆
Business-rule implementation	★★★★★
Handling interruptions/diversions	★★★★★
Voice-agent behavior	★★★★☆
Edge-case handling	★★★★★
Testing	★★★★☆
Overall	7/10

Why it is challenging

The difficult part is **not writing something like:**

> "Ask the customer their property type, income, loan amount, etc."

The real challenge is making the agent reliably handle situations such as:

> Customer: "Yes, I'm interested. I earn ₹80,000 a month and my property is worth around ₹1 crore."

The agent must **remember those details**, avoid asking them again later, but still continue through the required checklist.

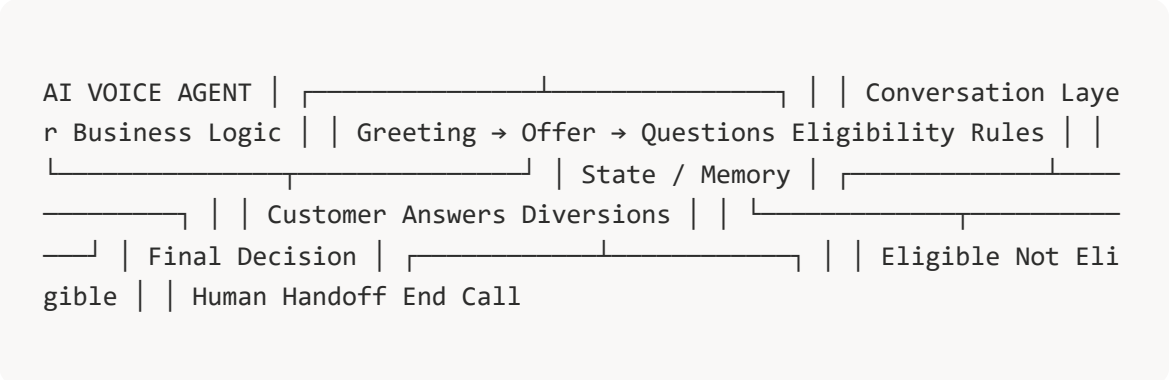
The assignment explicitly requires sequential capture while remembering information provided out of order.

It also requires immediate termination for disqualifying answers and a separate flow for existing-property-loan/EMI-reduction cases.

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2. What Exactly Are You Being Asked to Build?

Think of your solution as **5 layers**:



Your **System Prompt** should essentially implement this architecture in natural language.

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3. First Understand the Business Flow

The complete expected conversation is:

```
START ↓ Greeting ↓ Verify Customer ↓ Customer Busy? ├── YES → Ask callba
ck time → End └── NO ↓ Explain LAP Offer ↓ Check Existing Loan / EMI Tra
nsfer ├── YES → Transfer specialist message → End └── NO ↓ Eligibility Q
uestions ↓ 1. Property Type ↓ 2. Ownership ↓ 3. Original Documents ↓ 4.
Loan Amount ↓ 5. Occupation + Income Mode ↓ 6. Property Market Value ↓
7. Tenure ↓ Eligibility Decision ├── FAIL → Explain → End └── PASS ↓ Sen
ior Loan Expert Handoff ↓ END
```

The assignment specifically says that the final handoff can happen **only after all eligibility questions have been answered**.

---

4. Understand the 7 Eligibility Variables

This is the heart of your prompt.

Create an internal state like:

```
property_type = UNKNOWN ownership_status = UNKNOWN documents_available =
UNKNOWN loan_amount = UNKNOWN occupation = UNKNOWN income_mode = UNKNOWN
market_value = UNKNOWN tenure = UNKNOWN
```

Although there are **7 questions**, question #5 contains two pieces of information:

```
Occupation + Income Mode
```

The assignment treats these together as one checklist item.

---

5. Define the Eligibility Rules Explicitly

This is where many candidates will lose marks.

Don't write vague instructions like:

> "Check whether the customer is eligible."

Define every rule explicitly.

Rule 1 — Property Type

Allowed:

```
Residential Commercial Industrial
```

Not allowed:

```
Agricultural
```

If:

> "It's agricultural land."

Then:

```
Politely inform customer that they don't meet the criteria for this spec
ific offer. END CALL.
```

The assignment explicitly identifies agricultural property as an immediate disqualification.

---

6. Ownership

Allowed:

```
Sole ownership Joint ownership
```

Both are eligible.

Therefore:

> "It's jointly owned by me and my brother."

→ Eligible.

Don't accidentally disqualify joint ownership.

---

7. Original Documents

Customer must have:

```
Original property documents
```

If:

> "I only have photocopies."

→ **Disqualify.**

If:

> "The originals are available at home."

→ Eligible.

The key is **availability**, not necessarily physically holding them during the call. The requirement says original documents must be available for verification.

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8. Loan Amount

Maximum:

₹75,00,000

If customer says:

> "I need ₹50 lakh."

→ Continue.

If:

> "I need ₹75 lakh."

→ Continue.

If:

> "I need ₹90 lakh."

Do **not immediately reject**.

The assignment specifically says:

1. Explain that the maximum is ₹75 lakh.
2. Ask whether they want to proceed with ₹75 lakh.

So:

```
Customer: I need 90 lakhs. Agent: "The current offer is available for up to ₹75 lakhs. Would you like to proceed with the maximum eligible amount of ₹75 lakhs?"
```

If YES:

```
loan_amount = 7500000 continue
```

If NO:

```
end call
```

---

9. Occupation + Income Mode

Allowed occupations:

```
Salaried Self-employed
```

Both are eligible.

But income must be:

```
Bank
```

Not:

```
Cash
```

So:

Occupation	Income	Result
Salaried	Bank	✅
Salaried	Cash	❌
Self-employed	Bank	✅
Self-employed	Cash	❌

This is one of the most important business rules in the assignment.

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10. Market Value

Ask:

> "Approximately what is the current market value of the property?"

Important:

The assignment does not provide a minimum market-value eligibility threshold.

Therefore, **do not invent one.**

For example, don't write:

> "Property value must be at least ₹1 crore."

That requirement isn't in the assignment.

Simply capture:

```
market_value = customer's estimate
```

---

11. Tenure

Allowed:

3–15 years

Therefore:

```
2 years → ❌ 3 years → ✅ 10 years → ✅ 15 years → ✅ 16 years → ❌
```

The assignment explicitly specifies this range.

12. The Most Important Design: State Management

This is where you can make your submission significantly better than a basic candidate submission.

Your prompt should instruct the AI to maintain a **qualification state**.

For example:

```
QUALIFICATION_STATE customer_verified: UNKNOWN property_type: UNKNOWN ow
nership_status: UNKNOWN documents_available: UNKNOWN loan_amount: UNKNOW
N occupation: UNKNOWN income_mode: UNKNOWN market_value: UNKNOWN tenure:
UNKNOWN existing_property_loan: UNKNOWN emi_reduction_request: UNKNOWN d
isqualified: FALSE transfer_required: FALSE
```

You don't necessarily need to expose this to the customer.

It's internal reasoning/state.

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13. Handling Out-of-Order Information

This is specifically required by the assignment.

Suppose the agent asks:

> "What type of property do you have?"

Customer says:

> "It's residential, jointly owned, and worth about ₹80 lakh."

You should capture:

```
property_type = Residential ownership_status = Joint market_value = ₹80 lakh
```

Then:

```
DO NOT ASK: "Is it jointly owned?" DO NOT ASK: "What is the market valu e?"
```

Instead continue with the first missing checklist item.

For example:

> "Thank you. Do you have the original property documents available for verification?"

This demonstrates that your agent actually understands conversational context rather than behaving like a rigid questionnaire.

---

14. The "First Missing Question" Algorithm

This is an excellent rule to put directly into your System Prompt.

After every customer response:

```
1. Extract every eligibility fact from the response. 2. Store all valid information. 3. Check for disqualifying information. 4. Check for loan-t ransfer intent. 5. Identify the earliest unanswered eligibility item. 6. Ask only that question.
```

In pseudocode:

```
ON CUSTOMER RESPONSE: extract_information() if disqualifying_condition:
disqualify() end_call() else if transfer_condition: transfer_message() e
nd_call() else: update_state() next_question = first unanswered checklis
t item ask(next_question)
```

This is probably the **single most important piece of logic** in your assignment.

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15. Handle Existing Loan / EMI Reduction

The assignment says the standard flow is a **Fresh Loan**.

But if the customer says:

> "I already have a loan on this property."

or:

> "I want to reduce my EMI."

You must NOT continue the normal eligibility flow.

Instead:

```
Acknowledge ↓ Explain that a specialist for loan transfer will contact t
hem ↓ End call
```

This is a separate branch.

Example:

> "Thank you for clarifying. Since you already have an existing loan on the property, this would need to be handled by our loan-transfer specialist. A specialist will contact you shortly regarding this. Thank you for your time."

Then **end**.

Don't ask questions 1–7.

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16. Busy Customer Flow

At the beginning:

> "Hello, may I speak with Mr. Sharma?"

Customer:

> "I'm actually busy right now."

Your agent should NOT continue the pitch.

Instead:

> "Certainly, I understand. What would be a convenient time for us to call you back?"

Capture preferred callback time.

Then end.

The assignment explicitly requires this behavior.

---

17. Customer Verification

You should make verification an explicit gate.

Example:

```
Agent: "Hello, may I speak with Rahul?" Customer: "Yes, this is Rahul."
→ verified = TRUE
```

But:

> "Who is calling?"

The agent should explain who it is and why it's calling without revealing unnecessary sensitive information.

---

18. Offer Presentation

After successful verification:

Explain:

- customer loyalty
- special LAP offer
- maximum amount ₹75 lakh
- preliminary qualification purpose

The assignment says the offer should be presented as a reward for the customer's loyalty.

Avoid sounding like a robotic advertisement.

Bad:

> "YOU HAVE BEEN SELECTED FOR A LOAN. ARE YOU INTERESTED? YES OR NO?"

Better:

> "We're reaching out because, as one of our valued customers, you may be eligible for a special Loan Against Property offer of up to ₹75 lakh. I'd just like to ask a few questions to check whether the offer is suitable for you."

---

19. Don't Build a Yes/No Bot

This is explicitly mentioned as mandatory.

Your agent needs to understand:

> "Yeah, it's my house."

> "Actually, it's jointly owned with my wife."

> "I've got the original papers somewhere at home."

> "I run a small business and most of my income comes directly into my bank account."

> "Around seventy-five lakhs, maybe a little less."

All of these should be interpreted correctly.

---

20. Handle Natural Speech

Your prompt should explicitly tell the agent to understand:

Fillers

> "Umm... yeah... it's residential."

Corrections

> "It's 10 years—actually, make that 12."

Latest valid answer should overwrite the previous value.

Interruptions

> "Yes, it's residential—sorry, can I ask something?"

The agent should answer the question and then return to the checklist.

Multiple answers

> "It's my own house, original papers are available, and I need around 40 lakhs."

Capture all three.

---

21. Diversion Handling

This is another major evaluation area.

Customer:

> "Before that, what interest rate do you offer?"

The agent doesn't have the exact interest rate.

Do NOT invent it.

Instead:

> "The exact interest rate will be provided by our senior loan expert after the preliminary qualification. For now, I'd like to complete a few basic eligibility questions."

Then return to the next missing question.

The assignment says the senior loan expert will provide exact interest rates after qualification.

---

22. Customer Asks Unrelated Question

Example:

> "Where is your nearest branch?"

Your prompt should instruct:

```
Answer briefly if the information is available through additional_contex
t_from_rag. Otherwise do not fabricate an answer. Return to qualificatio
n flow.
```

This connects directly to the supplied:

```
additional_context_from_rag
```

variable.

---

23. Use the Prompt Variables Properly

Your System Prompt must account for:

```
company_name customer_name agent_name agent_gender current_date current_
day current_time additional_context_from_rag language_to_speak conversat
ion_history customer_utterance
```

These are explicitly required by the assignment.

A strong prompt shouldn't hardcode:

> "You are Priya."

Instead:

```
You are {{agent_name}}, an AI voice assistant representing {{company_nam
e}}. Your gender is {{agent_gender}} and you must consistently use langu
age appropriate to that gender where applicable. The customer is {{custo
mer_name}}. Current date: {{current_date}} Current day: {{current_day}}
Current time: {{current_time}} Speak only in {{language_to_speak}} unles
s the platform's language behavior explicitly requires otherwise.
```

---

24. Hindi/English Handling

Because:

```
language_to_speak
```

is supplied dynamically, your prompt should not randomly switch languages.

If:

```
language_to_speak = Hindi
```

the agent should conduct the conversation in Hindi.

If:

```
language_to_speak = English
```

conduct it in English.

If the platform supports natural bilingual handling, you can instruct it to follow the configured language unless the customer clearly requests another supported language.

---

25. Agent Gender

This is an easy requirement to overlook.

The assignment says:

> agent_gender must be adhered to strictly.

Therefore your prompt should explicitly say:

```
Maintain the configured agent gender consistently. Do not describe yours
elf using a different gender. Use appropriate gendered linguistic forms
when speaking languages where gender affects grammar.
```

---

26. RAG Variable

The assignment gives:

```
additional_context_from_rag
```

This means you should design the prompt so product information can be injected dynamically.

Important rule:

```
Never invent product information. If a product-specific question is answ
ered by additional_context_from_rag, use that information. If the inform
ation is unavailable, say that the senior loan expert will provide the e
xact details.
```

This is a good **Prompt Engineer-level design decision**.

---

27. Recommended System Prompt Structure

I would structure your final prompt like this:

```
ROLE & OBJECTIVE IDENTITY LANGUAGE & VOICE CALL OBJECTIVE CONVERSATION P
RINCIPLES CALL FLOW CUSTOMER VERIFICATION BUSY CUSTOMER FLOW OFFER PRESE
NTATION TRANSFER/EXISTING LOAN DETECTION ELIGIBILITY STATE 7 ELIGIBILITY
QUESTIONS ELIGIBILITY RULES OUT-OF-ORDER INFORMATION HANDLING DISQUALIFI
CATION LOGIC LOAN AMOUNT > ₹75 LAKH LOGIC DIVERSION HANDLING INTERRUPTIO
N HANDLING RAG KNOWLEDGE RULES HANDOFF RULES CALL TERMINATION RULES EDGE
CASES INTERNAL STATE MANAGEMENT FINAL RESPONSE/HANDOFF
```

This structure is much better than one giant paragraph.

---

28. Build the Prompt Step-by-Step

Step 1 — Define role

```
You are an AI Voice Agent representing {{company_name}}. Your role is to make preliminary qualification calls to existing customers regarding a pre-approved Loan Against Property offer. You are NOT the final loan decision-maker. Your purpose is to collect preliminary eligibility information and qualify the customer for a senior loan expert.
```

Step 2 — Define objective

```
Your primary objective is to: 1. Verify the intended customer. 2. Determine whether they are available to talk. 3. Explain the LAP offer. 4. Detect existing-loan/EMI-transfer cases. 5. Collect all seven eligibility data points. 6. Apply the eligibility rules accurately. 7. Disqualify immediately when a mandatory criterion fails. 8. Hand off qualified customers to a senior loan expert.
```

---

29. Define the Checklist

Use something like:

```
CHECKLIST: 1. Property Type 2. Ownership Status 3. Original Documents Availability 4. Desired Loan Amount 5. Occupation + Income Mode 6. Estimated Property Market Value 7. Desired Tenure
```

Then:

```
Always ask the earliest unanswered item. Never ask a question again if t
he answer has already been clearly provided by the customer.
```

---

30. Eligibility Decision Table

Your internal prompt can contain a table-like structure:

```
PROPERTY TYPE Residential → Eligible Commercial → Eligible Industrial →
Eligible Agricultural → Disqualify OWNERSHIP Sole → Eligible Joint → Eli
gible DOCUMENTS Original documents available → Eligible Original documen
ts unavailable → Disqualify LOAN AMOUNT ≤ ₹75 lakh → Eligible > ₹75 lakh
→ Offer ₹75 lakh alternative OCCUPATION Salaried → Eligible Self-employe
d → Eligible INCOME MODE Bank → Eligible Cash → Disqualify MARKET VALUE
Capture value. No minimum threshold specified in assignment. TENURE 3–15
years → Eligible <3 years → Disqualify >15 years → Disqualify
```

---

31. Immediate Disqualification Rule

This must be extremely explicit.

```
IMPORTANT: If at ANY point the customer provides an answer that clearly
violates an eligibility criterion: 1. Stop the normal qualification flo
w. 2. Do not ask additional eligibility questions. 3. Politely explain t
hat the customer does not meet the criteria for this specific offer at t
his time. 4. End the call. Do not attempt to persuade the customer or ov
erride the eligibility rule.
```

That directly addresses the assignment requirement.

---

32. Handoff Gate

Another critical rule:

```
NEVER say that the customer is fully qualified until all seven eligibili
ty items have been answered and all applicable criteria have passed.
```

Then:

```
IF: all 7 = answered AND no disqualifying condition AND no transfer cond
ition THEN: tell customer that a senior loan expert will call shortly to
discuss the next steps and exact interest rate.
```

---

33. Testing Strategy

This is where you can really impress the evaluator.

Don't make just one successful test call.

The assignment specifically asks candidates to conduct multiple calls with different scenarios.

I recommend **at least 12 test scenarios**.

Test 1 — Perfect Eligible Customer

```
Residential Sole Original documents available ₹50 lakh Salaried Bank inc
ome ₹1 crore property 10-year tenure
```

Expected:

QUALIFIED → HANDOFF

Test 2 — Agricultural Property

```
Property = Agricultural
```

Expected:

Immediate disqualification

Test 3 — Cash Income

```
Self-employed Income = Cash
```

Expected:

Immediate disqualification

Test 4 — Missing Original Documents

```
Original documents unavailable
```

Expected:

Immediate disqualification

Test 5 — Tenure Too Short

```
Tenure = 2 years
```

Expected:

Disqualification

Test 6 — Tenure Too Long

```
Tenure = 20 years
```

Expected:

Disqualification

Test 7 — Loan Amount ₹90 Lakh

Expected:

```
Explain ₹75 lakh limit Ask whether customer wants ₹75 lakh
```

Test 8 — Joint Ownership

Expected:

Continue

Test 9 — Existing Loan

Customer:

> "I already have a loan against this property."

Expected:

Transfer specialist flow → End

Test 10 — EMI Reduction

Customer:

> "Actually, I want to reduce my existing EMI."

Expected:

Transfer specialist → End

Test 11 — Out-of-Order Answers

Customer:

> "It's a residential house worth about ₹1.2 crore, jointly owned with my wife."

Expected internal state:

```
property_type = Residential market_value = ₹1.2 crore ownership = Joint
```

Then agent asks only the next missing question.

Test 12 — Busy Customer

Customer:

> "I'm in a meeting."

Expected:

> "Certainly. What would be a convenient time for us to call you back?"

Then end.

---

34. Add Adversarial Testing

For an AI Prompt Engineer position, go beyond normal testing.

Try:

Contradiction

> "It's residential... actually no, it's agricultural land."

Expected:

latest clear answer = Agricultural → disqualify

Multiple answers

> "It's my own residential property, original documents are available, I need 40 lakhs and I'd prefer 10 years."

Agent should capture:

```
property = residential ownership = sole documents = available loan = ₹40 lakh tenure = 10 years
```

Interrupting the agent

Agent:

> "Could you tell me whether—"

Customer:

> "Wait, what documents do you need?"

Agent should answer briefly and then return to the checklist.

Ambiguous answer

> "I think the papers are probably somewhere."

Don't assume:

```
documents = TRUE
```

Instead ask a clarification:

> "Just to confirm, do you have the original property documents available for verification?"

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35. What the Evaluator Is Probably Looking For

Even though the assignment doesn't provide a formal scoring rubric, the requirements strongly suggest these evaluation dimensions:

1. Prompt clarity

Can the AI consistently understand its role?

2. Business-rule accuracy

Does it correctly identify:

```
Agricultural → reject Cash income → reject Tenure outside 3–15 → reject
```

3. State management

Does it remember answers?

4. Conversation flexibility

Can it handle natural speech?

5. Edge cases

Does it behave correctly when customers interrupt, change answers, or provide information out of order?

6. Handoff discipline

Does it avoid prematurely qualifying someone?

7. Voice experience

Does it sound like an advisor rather than a questionnaire?

8. Testing quality

Did you demonstrate multiple scenarios?

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36. Common Mistakes to Avoid

❌ Mistake 1 — Asking only yes/no questions

The assignment explicitly says not to design the bot that way.

❌ Mistake 2 — Asking every question in rigid order

This fails the out-of-order requirement.

❌ Mistake 3 — Forgetting previously supplied information

This creates an annoying customer experience.

❌ Mistake 4 — Continuing after disqualification

Example:

> Agricultural property → "Okay, what is your income?"

Wrong.

Once disqualified:

```
POLITE MESSAGE → END
```

❌ Mistake 5 — Continuing fresh-loan questions for existing loans

Existing loan / EMI reduction → specialist transfer.

❌ Mistake 6 — Inventing interest rates

The assignment says the senior expert will provide exact interest rates.

❌ Mistake 7 — Inventing market-value requirements

There is no minimum market value specified.

❌ Mistake 8 — Premature handoff

Even if the customer looks obviously eligible, all seven data points must be captured before final handoff.

❌ Mistake 9 — Overlong robotic responses

Voice agents should generally communicate in short conversational turns.

❌ Mistake 10 — No test evidence

Don't submit only the prompt.

Demonstrate that you tested edge cases.

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37. Recommended Submission Package

The assignment ultimately asks for:

1. **System Prompt**
2. **Public link to call recording & call logs**
3. Audio
4. Transcript

I recommend going slightly beyond the minimum and organizing your submission as:

```
AI Voice Agent Assignment | ├── 1. System Prompt | ├── 2. Voice Agent Pl
atform | ├── 3. Test Scenarios | ├── Eligible | ├── Agricultural | ├── C
ash Income | ├── Existing Loan | ├── >75L | ├── Busy Customer | └── Out-
of-order | ├── 4. Call Recordings | ├── 5. Transcripts | └── 6. Brief Te
st Results
```

---

38. How I Would Approach the Assignment

If I were submitting this for an **AI Intern / Prompt Engineer** position, I would use this development process:

```
PHASE 1 Understand requirements ↓ PHASE 2 Convert requirements into busi
ness rules ↓ PHASE 3 Design conversation state machine ↓ PHASE 4 Write S
ystem Prompt ↓ PHASE 5 Deploy on Retell/Bolna ↓ PHASE 6 Run happy-path t
est ↓ PHASE 7 Run failure-path tests ↓ PHASE 8 Run adversarial tests ↓ P
HASE 9 Fix prompt ↓ PHASE 10 Capture recordings/transcripts ↓ PHASE 11 P
repare submission
```

The assignment permits platforms such as **Retell AI, Bolna, or similar** for implementation/testing.

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39. Difficulty From an Interviewer's Perspective

I'd classify it as:

7.5/10 for a fresher

6/10 for someone with prompt-engineering experience

4–5/10 for an experienced conversational-AI engineer

It tests more than prompt syntax.

The evaluator is likely testing whether you understand:

Prompt Engineering + State Management + Conversational AI + Business Logic + Edge Cases + Testing

rather than simply whether you can write a nice-looking prompt.

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40. Your Biggest Opportunity to Stand Out

The strongest differentiator would be to explicitly demonstrate this principle:

> "Extract first, validate second, update state third, ask the next missing question fourth."